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Институт №8 “Компьютерные науки и прикладная
математика” Кафедра №806 “Вычислительная математика и
программирование”

Лабораторная работа №2 по курсу

«Операционные системы»

Группа: М8О-211Б-23

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Оценка: _____

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Постановка задачи

Цель работы:

Целью является приобретение практических навыков в управление потоками в ОС и обеспечение синхронизации между потоками.

Задание:

Составить программу на языке Си, обрабатывающую данные в многопоточном режиме. При обработке использовать стандартные средства создания потоков операционной системы (Windows/Unix). Ограничение максимального количества потоков, работающих в один момент времени, должно быть задано ключом запуска вашей программы. Так же необходимо уметь продемонстрировать количество потоков, используемое вашей программой с помощью стандартных средств операционной системы. В отчете привести исследование зависимости ускорения и эффективности алгоритма от входных данных и количества потоков. Получившиеся результаты необходимо объяснить.

Вариант 4.

Отсортировать массив целых чисел при помощи сортировки слиянием.

Общий метод и алгоритм решения

Использованные системные вызовы:

- `int pthread_create(pthread_t * __restrict, const pthread_attr_t * __restrict, void *(* __start_routine)(void *), void * __restrict, void * __restrict)` — создаёт поток с рутинной (стартовой функцией) и заданными аргументами
- `int pthread_join(pthread_t __th, void **_thread_return)` — дожидается завершения потока (Эта процедура блокирует вызывающий поток до тех пор, пока не будет

осуществлен выход из другого (указанного) потока).

Для разграничения доступа потоков к общим ресурсам могут использоваться мьютексы. Мьютекс представляет объект, который может находиться в двух состояниях: заблокированном и разблокированном. На уровне языка Си мьютекс представлен типом `pthread_mutex_t`.

`int pthread_mutex_init(pthread_mutex_t *mutex, const pthread_mutexattr_t *mutexattr)` – инициализация мьютекса;

`int pthread_mutex_lock(pthread_mutex_t *mutex)` – блокировка мьютекса;

`int pthread_mutex_unlock(pthread_mutex_t *mutex)` – разблокировка мьютекса;

`int pthread_mutex_destroy(pthread_mutex_t *mutex)` – уничтожение мьютекса;

Программа получает на вход два аргумента – размер массива и максимально количество потоков для сортировки. Для полученного размера массива, происходит его заполнение случайными значениями с помощью функции `srand()`.

Мьютекс используется для синхронизации доступа к глобальной переменной `thread_count`, отслеживающей общее количество созданных потоков.

Сортировка сливанием

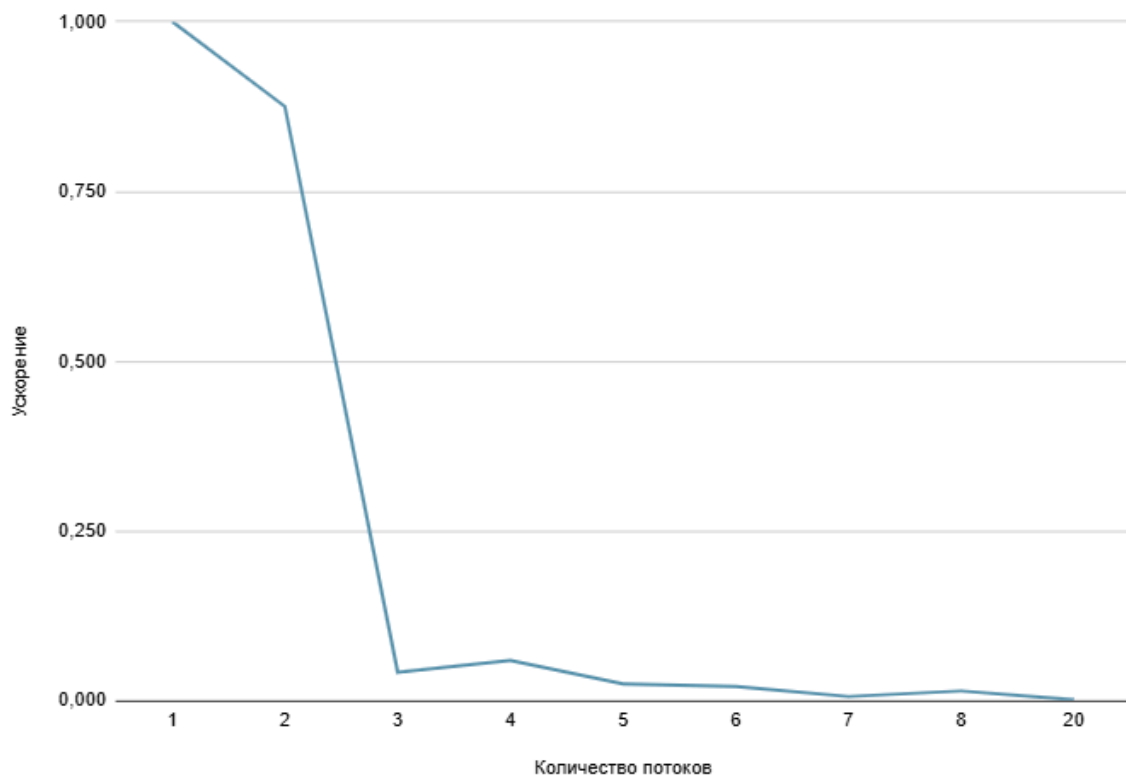
Разбиваем текущий массив пополам, сливаем левый и правый подмассив. По правилу: если элемент из левого подмассива меньше элемента из правого подмассива записываем левый в результирующий массив и увеличиваем индекс, отвечающий за начало левого массива, иначе записываем элемент из правого подмассива и сдвигаем указатель, указывающий на начало правого массива. Сортировку производим рекурсивно. При рекурсии создаем новые потоки, чтобы левая и правая части массива сортировались параллельно.

Ниже приведены данные, показывающие изменение ускорения и эффективности, с разным количеством потоков, для этой реализации. Размер массива 1000.

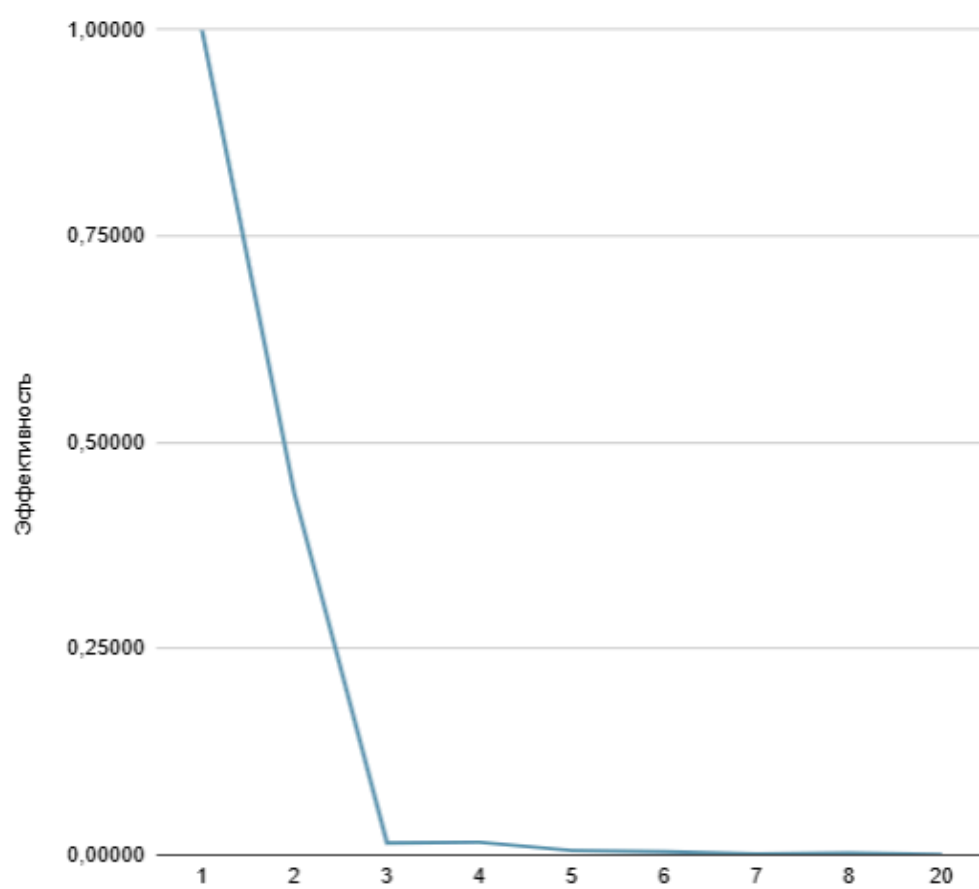
Число потоков	Время выполнения (с)	Ускорение	Эффективность
1	0,000007	1,000	1,00000
2	0,000008	0,875	0,43750
3	0,000167	0,042	0,01397
4	0,000118	0,059	0,01483
5	0,000282	0,025	0,00496
6	0,000332	0,021	0,00351
7	0,001129	0,006	0,00089
8	0,000481	0,015	0,00182
20	0,004878	0,001	0,00007

Размер массива	Время выполнения(с)
2	0.000002
4	0.000001
100	0.000002
1000	0.000007
15000	0.000093
150001	0.009052
100000	0.000621
1000000	0.027520

Зависимость ускорения от количества потоков



Зависимость эффективности от количества потоков



Код программы

```
#include <stdio.h>
#include <stdlib.h>
#include <pthread.h>
#include <time.h>

typedef struct
{
    int *array;
    int left;
    int right;
    int cnt;
} MergeSortArgs;

pthread_mutex_t mutex; // Мьютекс для синхронизации
int thread_count = 0; // Количество активных потоков

void merge(int *array, int left, int mid, int right)
{
    int i, j, k;
    int n1 = mid - left + 1;
    int n2 = right - mid;

    int *L = (int *)malloc(n1 * sizeof(int));
    int *R = (int *)malloc(n2 * sizeof(int));

    for (i = 0; i < n1; i++)
        L[i] = array[left + i];
    for (j = 0; j < n2; j++)
```

```
R[j] = array[mid + 1 + j];
```

```
i = 0;
```

```
j = 0;
```

```
k = left;
```

```
while (i < n1 && j < n2)
```

```
{
```

```
    if (L[i] <= R[j])
```

```
        array[k++] = L[i++];
```

```
    else
```

```
        array[k++] = R[j++];
```

```
}
```

```
while (i < n1)
```

```
    array[k++] = L[i++];
```

```
while (j < n2)
```

```
    array[k++] = R[j++];
```

```
free(L);
```

```
free(R);
```

```
}
```

```
void *merge_sort(void *args)
```

```
{
```

```
    MergeSortArgs *ms_args = (MergeSortArgs *)args;
```

```
    int left = ms_args->left;
```

```
    int right = ms_args->right;
```

```
    int mid;
```

```
    int cnt = ms_args->cnt;
```

```
pthread_mutex_lock(&mutex);
thread_count += 2; // Увеличение счетчика активных потоков
pthread_mutex_unlock(&mutex);

if (left < right)
{
    mid = left + (right - left) / 2;

    MergeSortArgs left_args = {ms_args->array, left, mid, cnt};
    MergeSortArgs right_args = {ms_args->array, mid + 1, right, cnt};

    pthread_t thread1, thread2;

    if (thread_count < cnt)
    {
        pthread_create(&thread1, NULL, merge_sort, &left_args);
        pthread_create(&thread2, NULL, merge_sort, &right_args);

        pthread_join(thread1, NULL);
        pthread_join(thread2, NULL);
    }

    merge(ms_args->array, left, mid, right);
}

pthread_mutex_lock(&mutex);
--thread_count; // Уменьшение счетчика активных потоков
pthread_mutex_unlock(&mutex);
```

```

    return NULL;
}

int main(int argc, char *argv[])
{
    if (argc != 3)
    {
        printf("Usage: %s <array_size> <num_threads>\n", argv[0]);
        return 1;
    }

    int n = atoi(argv[1]);
    int num_threads = atoi(argv[2]);

    int *arr = (int *)malloc(n * sizeof(int));
    srand(100);
    for (int i = 0; i < n; i++)
    {
        arr[i] = rand() % 100;
    }

    pthread_mutex_init(&mutex, NULL);
    MergeSortArgs args = {arr, 0, n - 1, num_threads};

    clock_t start_time = clock();
    merge_sort(&args);
    clock_t end_time = clock();

    double time_spent = (double)(end_time - start_time) / CLOCKS_PER_SEC;
    ++thread_count;

```

```
printf("The value of the requested threads: %d\n", num_threads);  
printf("The total number of threads created: %d\n", thread_count);  
printf("Elapsed time: %f seconds\n", time_spent);
```

```
free(arr);
```

```
pthread_mutex_destroy(&mutex);
```

```
return 0;
```

```
}
```

Протокол работы программы

leoelena@DESKTOP-HJEL67G:/mnt/c/Users/Елена/Desktop/OS/OS/LAB_2/src\$./main
1024 8

Number of threads requested: 8

Total threads created: 7

Time taken: 0.002196 seconds

leoelena@DESKTOP-HJEL67G:/mnt/c/Users/Елена/Desktop/OS/OS/LAB_2/src\$./main
1024 20

Number of threads requested: 20

Total threads created: 15

Time taken: 0.013289 seconds

leoelena@DESKTOP-HJEL67G:/mnt/c/Users/Елена/Desktop/OS/OS/LAB_2/src\$./main
1024 7

Number of threads requested: 7

Total threads created: 7

Time taken: 0.005926 seconds

leoelena@DESKTOP-HJEL67G:/mnt/c/Users/Елена/Desktop/OS/OS/LAB_2/src\$./main
1024 1

Number of threads requested: 1

Total threads created: 1

Time taken: 0.001070 seconds

leoelena@DESKTOP-HJEL67G:/mnt/c/Users/Елена/Desktop/OS/OS/LAB_2/src\$./main
16 5

Number of threads requested: 5

Total threads created: 3

Time taken: 0.000874 seconds

leoelena@DESKTOP-HJEL67G:/mnt/c/Users/Елена/Desktop/OS/OS/LAB_2/src\$./main
16 3

Number of threads requested: 3

Total threads created: 3

Time taken: 0.001546 seconds

leoelena@DESKTOP-HJEL67G:/mnt/c/Users/Елена/Desktop/OS/OS/LAB_2/src\$./main
20 4

Array size must be a power of 2

leoelena@DESKTOP-HJEL67G:/mnt/c/Users/Елена/Desktop/OS/OS/LAB_2/src\$ strace
-f ./main 1024 4

execve("./main", ["./main", "1024", "4"], 0x7ffd83a382b8 /* 36 vars */) = 0

brk(NULL) = 0x5580da4d4000

mmap(NULL, 8192, PROT_READ|PROT_WRITE,
MAP_PRIVATE|MAP_ANONYMOUS, -1, 0) = 0x7fa97868d000

access("/etc/ld.so.preload", R_OK) = -1 ENOENT (No such file or directory)

openat(AT_FDCWD, "/etc/ld.so.cache", O_RDONLY|O_CLOEXEC) = 3

fstat(3, {st_mode=S_IFREG|0644, st_size=20115, ...}) = 0

mmap(NULL, 20115, PROT_READ, MAP_PRIVATE, 3, 0) = 0x7fa978688000

close(3) = 0

openat(AT_FDCWD, "/lib/x86_64-linux-gnu/libtsan.so.2", O_RDONLY|O_CLOEXEC) =
3

read(3, "\177ELF\2\1\1\0\0\0\0\0\0\0\0\0\3\0>\0\1\0\0\0\0\0\0\0\0\0\0\0"..., 832) = 832

fstat(3, {st_mode=S_IFREG|0644, st_size=8980352, ...}) = 0

mmap(NULL, 17393288, PROT_READ, MAP_PRIVATE|MAP_DENYWRITE, 3, 0) =
0x7fa9775f1000

mmap(0x7fa97762b000, 856064, PROT_READ|PROT_EXEC,
MAP_PRIVATE|MAP_FIXED|MAP_DENYWRITE, 3, 0x3a000) = 0x7fa97762b000

mmap(0x7fa9776fc000, 212992, PROT_READ,
MAP_PRIVATE|MAP_FIXED|MAP_DENYWRITE, 3, 0x10b000) = 0x7fa9776fc000

mmap(0x7fa977730000, 49152, PROT_READ|PROT_WRITE,
MAP_PRIVATE|MAP_FIXED|MAP_DENYWRITE, 3, 0x13e000) = 0x7fa977730000

mmap(0x7fa97773c000, 16037512, PROT_READ|PROT_WRITE,

```

MAP_PRIVATE|MAP_FIXED|MAP_ANONYMOUS, -1, 0) = 0x7fa97773c000
close(3) = 0
openat(AT_FDCWD, "/lib/x86_64-linux-gnu/libc.so.6", O_RDONLY|O_CLOEXEC) = 3
read(3, "\177ELF\2\1\1\3\0\0\0\0\0\0\0\3\0>\0\1\0\0\0\220\243\2\0\0\0\0\0"..., 832) =
832
pread64(3, "\6\0\0\0\4\0\0\0@\0\0\0\0\0\0\0@\0\0\0\0\0\0\0@\0\0\0\0\0\0\0"..., 784, 64) =
784
fstat(3, {st_mode=S_IFREG|0755, st_size=2125328, ...}) = 0
pread64(3, "\6\0\0\0\4\0\0\0@\0\0\0\0\0\0\0@\0\0\0\0\0\0\0@\0\0\0\0\0\0\0"..., 784, 64) =
784
mmap(NULL, 2170256, PROT_READ, MAP_PRIVATE|MAP_DENYWRITE, 3, 0) =
0x7fa97773df000
mmap(0x7fa977407000, 1605632, PROT_READ|PROT_EXEC,
MAP_PRIVATE|MAP_FIXED|MAP_DENYWRITE, 3, 0x28000) = 0x7fa977407000
mmap(0x7fa97758f000, 323584, PROT_READ,
MAP_PRIVATE|MAP_FIXED|MAP_DENYWRITE, 3, 0x1b0000) = 0x7fa97758f000
mmap(0x7fa9775de000, 24576, PROT_READ|PROT_WRITE,
MAP_PRIVATE|MAP_FIXED|MAP_DENYWRITE, 3, 0x1fe000) = 0x7fa9775de000
mmap(0x7fa9775e4000, 52624, PROT_READ|PROT_WRITE,
MAP_PRIVATE|MAP_FIXED|MAP_ANONYMOUS, -1, 0) = 0x7fa9775e4000
close(3) = 0
openat(AT_FDCWD, "/lib/x86_64-linux-gnu/libm.so.6", O_RDONLY|O_CLOEXEC) = 3
read(3, "\177ELF\2\1\1\3\0\0\0\0\0\0\0\3\0>\0\1\0\0\0\0\0\0\0\0\0\0\0\0\0"..., 832) = 832
fstat(3, {st_mode=S_IFREG|0644, st_size=952616, ...}) = 0
mmap(NULL, 950296, PROT_READ, MAP_PRIVATE|MAP_DENYWRITE, 3, 0) =
0x7fa97772f6000
mmap(0x7fa977306000, 520192, PROT_READ|PROT_EXEC,
MAP_PRIVATE|MAP_FIXED|MAP_DENYWRITE, 3, 0x10000) = 0x7fa977306000
mmap(0x7fa977385000, 360448, PROT_READ,
MAP_PRIVATE|MAP_FIXED|MAP_DENYWRITE, 3, 0x8f000) = 0x7fa977385000

```

```

mmap(0x7fa9773dd000, 8192, PROT_READ|PROT_WRITE,
MAP_PRIVATE|MAP_FIXED|MAP_DENYWRITE, 3, 0xe7000) = 0x7fa9773dd000
close(3) = 0
openat(AT_FDCWD, "/lib/x86_64-linux-gnu/libgcc_s.so.1", O_RDONLY|O_CLOEXEC)
= 3
read(3, "\177ELF\2\1\1\0\0\0\0\0\0\0\0\0\3\0>\0\1\0\0\0\0\0\0\0\0\0\0"..., 832) = 832
fstat(3, {st_mode=S_IFREG|0644, st_size=183024, ...}) = 0
mmap(NULL, 185256, PROT_READ, MAP_PRIVATE|MAP_DENYWRITE, 3, 0) =
0x7fa9772c8000
mmap(0x7fa9772cc000, 147456, PROT_READ|PROT_EXEC,
MAP_PRIVATE|MAP_FIXED|MAP_DENYWRITE, 3, 0x4000) = 0x7fa9772cc000
mmap(0x7fa9772f0000, 16384, PROT_READ,
MAP_PRIVATE|MAP_FIXED|MAP_DENYWRITE, 3, 0x28000) = 0x7fa9772f0000
mmap(0x7fa9772f4000, 8192, PROT_READ|PROT_WRITE,
MAP_PRIVATE|MAP_FIXED|MAP_DENYWRITE, 3, 0x2b000) = 0x7fa9772f4000
close(3) = 0
mmap(NULL, 8192, PROT_READ|PROT_WRITE,
MAP_PRIVATE|MAP_ANONYMOUS, -1, 0) = 0x7fa9772c6000
mmap(NULL, 12288, PROT_READ|PROT_WRITE,
MAP_PRIVATE|MAP_ANONYMOUS, -1, 0) = 0x7fa9772c3000
arch_prctl(ARCH_SET_FS, 0x7fa9772c3f00) = 0
set_tid_address(0x7fa9772c41d0) = 63073
set_robust_list(0x7fa9772c41e0, 24) = 0
rseq(0x7fa9772c4820, 0x20, 0, 0x53053053) = 0
mprotect(0x7fa9775de000, 16384, PROT_READ) = 0
mprotect(0x7fa9772f4000, 4096, PROT_READ) = 0
mprotect(0x7fa9773dd000, 4096, PROT_READ) = 0
mprotect(0x7fa977730000, 16384, PROT_READ) = 0
mprotect(0x5580d9cfc000, 4096, PROT_READ) = 0
mprotect(0x7fa9786c5000, 8192, PROT_READ) = 0

```

```
prlimit64(0, RLIMIT_STACK, NULL, {rlim_cur=8192*1024,
rlim_max=RLIM64_INFINITY}) = 0
munmap(0x7fa978688000, 20115)      = 0
mmap(NULL, 4096, PROT_READ|PROT_WRITE,
MAP_PRIVATE|MAP_ANONYMOUS, -1, 0) = 0x7fa97868c000
mmap(NULL, 4096, PROT_READ|PROT_WRITE,
MAP_PRIVATE|MAP_ANONYMOUS, -1, 0) = 0x7fa97868b000
openat(AT_FDCWD, "/proc/self/environ", O_RDONLY) = 3
read(3, "SHELL=/bin/bash\0COLORTERM=trueco"..., 4096) = 4096
close(3)                                = 0
munmap(0x7fa97868b000, 4096)      = 0
mmap(NULL, 8192, PROT_READ|PROT_WRITE,
MAP_PRIVATE|MAP_ANONYMOUS, -1, 0) = 0x7fa97868a000
openat(AT_FDCWD, "/proc/self/environ", O_RDONLY) = 3
read(3, "SHELL=/bin/bash\0COLORTERM=trueco"..., 8192) = 4725
read(3, "", 3467)                   = 0
close(3)                                = 0
readlinkat(AT_FDCWD, "/proc/self/exe",
"/mnt/c/Users/^320\225\320\273\320\265\320\275\320\260/Desktop/"..., 4096) = 52
mmap(NULL, 4096, PROT_READ|PROT_WRITE,
MAP_PRIVATE|MAP_ANONYMOUS, -1, 0) = 0x7fa978689000
openat(AT_FDCWD, "/proc/self/cmdline", O_RDONLY) = 3
read(3, "./main\0001024\0004\0", 4096) = 14
read(3, "", 4082)                   = 0
close(3)                                = 0
munmap(0x7fa978689000, 4096)      = 0
mmap(NULL, 65536, PROT_READ|PROT_WRITE,
MAP_PRIVATE|MAP_ANONYMOUS, -1, 0) = 0x7fa9772b3000
mmap(0x720000000000, 1099511635968, PROT_NONE,
MAP_PRIVATE|MAP_FIXED|MAP_ANONYMOUS|MAP_NORESERVE, -1, 0) =
```

0x720000000000

```
mmap(0x730000000000, 8192, PROT_READ|PROT_WRITE,  
MAP_PRIVATE|MAP_FIXED|MAP_ANONYMOUS, -1, 0) = 0x730000000000  
mmap(NULL, 8388608, PROT_NONE,  
MAP_PRIVATE|MAP_ANONYMOUS|MAP_NORESERVE, -1, 0) = 0x7fa976ab3000  
mmap(NULL, 4263936, PROT_READ|PROT_WRITE,  
MAP_PRIVATE|MAP_ANONYMOUS, -1, 0) = 0x7fa9766a2000  
mmap(NULL, 2097152, PROT_READ|PROT_WRITE,  
MAP_PRIVATE|MAP_ANONYMOUS, -1, 0) = 0x7fa9764a2000  
munmap(0x7fa9764a2000, 385024)      = 0  
munmap(0x7fa976600000, 663552)      = 0  
mmap(NULL, 4096, PROT_READ|PROT_WRITE,  
MAP_PRIVATE|MAP_ANONYMOUS, -1, 0) = 0x7fa978689000  
mmap(NULL, 3727360, PROT_READ|PROT_WRITE,  
MAP_PRIVATE|MAP_ANONYMOUS, -1, 0) = 0x7fa976172000  
clock_gettime(CLOCK_MONOTONIC, {tv_sec=14139, tv_nsec=102249214}) = 0  
mmap(0x721400000000, 65536, PROT_READ|PROT_WRITE,  
MAP_PRIVATE|MAP_FIXED|MAP_ANONYMOUS, -1, 0) = 0x721400000000  
mmap(0x721780000000, 65536, PROT_READ|PROT_WRITE,  
MAP_PRIVATE|MAP_FIXED|MAP_ANONYMOUS, -1, 0) = 0x721780000000  
clock_gettime(CLOCK_MONOTONIC, {tv_sec=14139, tv_nsec=102688516}) = 0  
mmap(0x720800000000, 65536, PROT_READ|PROT_WRITE,  
MAP_PRIVATE|MAP_FIXED|MAP_ANONYMOUS, -1, 0) = 0x720800000000  
mmap(0x720b80000000, 65536, PROT_READ|PROT_WRITE,  
MAP_PRIVATE|MAP_FIXED|MAP_ANONYMOUS, -1, 0) = 0x720b80000000  
clock_gettime(CLOCK_MONOTONIC, {tv_sec=14139, tv_nsec=103168105}) = 0  
mmap(0x721000000000, 65536, PROT_READ|PROT_WRITE,  
MAP_PRIVATE|MAP_FIXED|MAP_ANONYMOUS, -1, 0) = 0x721000000000  
mmap(0x721380000000, 65536, PROT_READ|PROT_WRITE,  
MAP_PRIVATE|MAP_FIXED|MAP_ANONYMOUS, -1, 0) = 0x721380000000
```

```
clock_gettime(CLOCK_MONOTONIC, {tv_sec=14139, tv_nsec=103611967}) = 0
mmap(0x721800000000, 65536, PROT_READ|PROT_WRITE,
MAP_PRIVATE|MAP_FIXED|MAP_ANONYMOUS, -1, 0) = 0x721800000000
mmap(0x721b80000000, 65536, PROT_READ|PROT_WRITE,
MAP_PRIVATE|MAP_FIXED|MAP_ANONYMOUS, -1, 0) = 0x721b80000000
mmap(NULL, 4096, PROT_READ|PROT_WRITE,
MAP_PRIVATE|MAP_ANONYMOUS, -1, 0) = 0x7fa978688000
prlimit64(0, RLIMIT_CORE, NULL, {rlim_cur=0, rlim_max=RLIM64_INFINITY}) = 0
prlimit64(0, RLIMIT_CORE, {rlim_cur=0, rlim_max=RLIM64_INFINITY}, NULL) = 0
prlimit64(0, RLIMIT_STACK, NULL, {rlim_cur=8192*1024,
rlim_max=RLIM64_INFINITY}) = 0
prlimit64(0, RLIMIT_AS, NULL, {rlim_cur=RLIM64_INFINITY,
rlim_max=RLIM64_INFINITY}) = 0
mmap(NULL, 4096, PROT_READ|PROT_WRITE,
MAP_PRIVATE|MAP_ANONYMOUS, -1, 0) = 0x7fa9766a1000
openat(AT_FDCWD, "/proc/self/maps", O_RDONLY) = 3
read(3, "5580d9cf9000-5580d9cfa000 r--p 0"..., 4096) = 4090
read(3, "7fa978", 6) = 6
close(3) = 0
munmap(0x7fa9766a1000, 4096) = 0
mmap(NULL, 8192, PROT_READ|PROT_WRITE,
MAP_PRIVATE|MAP_ANONYMOUS, -1, 0) = 0x7fa9766a0000
openat(AT_FDCWD, "/proc/self/maps", O_RDONLY) = 3
read(3, "5580d9cf9000-5580d9cfa000 r--p 0"..., 8192) = 4090
read(3, "7fa97868f000-7fa978690000 r--p 0"..., 4102) = 841
read(3, "", 3261) = 0
close(3) = 0
mmap(NULL, 4096, PROT_READ|PROT_WRITE,
MAP_PRIVATE|MAP_ANONYMOUS, -1, 0) = 0x7fa97669f000
openat(AT_FDCWD, "/proc/self/maps", O_RDONLY) = 3
```

```
read(3, "5580d9cf9000-5580d9cfa000 r--p 0"..., 4096) = 4090
read(3, "7fa978", 6) = 6
close(3) = 0
munmap(0x7fa97669f000, 4096) = 0
mmap(NULL, 8192, PROT_READ|PROT_WRITE,
MAP_PRIVATE|MAP_ANONYMOUS, -1, 0) = 0x7fa97669e000
openat(AT_FDCWD, "/proc/self/maps", O_RDONLY) = 3
read(3, "5580d9cf9000-5580d9cfa000 r--p 0"..., 8192) = 4090
read(3, "7fa97868f000-7fa978690000 r--p 0"..., 4102) = 841
read(3, "", 3261) = 0
close(3) = 0
mmap(0x200000000000, 15393162788864, PROT_NONE,
MAP_PRIVATE|MAP_FIXED|MAP_ANONYMOUS|MAP_NORESERVE, -1, 0) =
0x200000000000
mmap(0x380000000000, 31885837205504, PROT_NONE,
MAP_PRIVATE|MAP_FIXED|MAP_ANONYMOUS|MAP_NORESERVE, -1, 0) =
0x380000000000
mmap(0x5a0000000000, 26388279066624, PROT_NONE,
MAP_PRIVATE|MAP_FIXED|MAP_ANONYMOUS|MAP_NORESERVE, -1, 0) =
0x5a0000000000
mmap(0x730000002000, 7696581386240, PROT_NONE,
MAP_PRIVATE|MAP_FIXED|MAP_ANONYMOUS|MAP_NORESERVE, -1, 0) =
0x730000002000
munmap(0x7fa97669e000, 8192) = 0
mmap(0x100000000000, 35184372088832, PROT_READ|PROT_WRITE,
MAP_PRIVATE|MAP_FIXED|MAP_ANONYMOUS|MAP_NORESERVE, -1, 0) =
0x100000000000
madvise(0x100000000000, 35184372088832, MADV_NOHUGEPAGE) = 0
madvise(0x100000000000, 35184372088832, MADV_DONTDUMP) = 0
mmap(0x300000000000, 8796093022208, PROT_READ|PROT_WRITE,
```

```
MAP_PRIVATE|MAP_FIXED|MAP_ANONYMOUS|MAP_NORESERVE, -1, 0) =
0x300000000000
madvise(0x300000000000, 8796093022208, MADV_NOHUGEPAGE) = 0
madvise(0x300000000000, 8796093022208, MADV_DONTDUMP) = 0
getpid() = 63073
openat(AT_FDCWD, "/tmp/tsan.rodata.63073", O_RDWR|O_CREAT|O_EXCL, 0600) =
3
unlinkat(AT_FDCWD, "/tmp/tsan.rodata.63073", 0) = 0
mmap(NULL, 524288, PROT_READ|PROT_WRITE,
MAP_PRIVATE|MAP_ANONYMOUS, -1, 0) = 0x7fa976620000
write(3, "\0\0\0@\0\0\0@\0\0\0@\0\0\0@\0\0\0@\0\0\0@\0\0\0@\0\0\0@" ..., 524288) =
524288
mmap(NULL, 4096, PROT_READ|PROT_WRITE,
MAP_PRIVATE|MAP_ANONYMOUS, 3, 0) = 0x7fa97661f000
mmap(NULL, 4096, PROT_READ|PROT_WRITE,
MAP_PRIVATE|MAP_ANONYMOUS, -1, 0) = 0x7fa97661e000
openat(AT_FDCWD, "/proc/self/maps", O_RDONLY) = 4
read(4, "20000000000-100000000000 ---p 00" ..., 4096) = 4071
read(4, "7fa977734000-7fa97773c000", 25) = 25
close(4) = 0
munmap(0x7fa97661e000, 4096) = 0
mmap(NULL, 8192, PROT_READ|PROT_WRITE,
MAP_PRIVATE|MAP_ANONYMOUS, -1, 0) = 0x7fa97661d000
openat(AT_FDCWD, "/proc/self/maps", O_RDONLY) = 4
read(4, "20000000000-100000000000 ---p 00" ..., 8192) = 4071
read(4, "7fa977734000-7fa97773c000 rw-p 0" ..., 4121) = 1055
read(4, "", 3066) = 0
close(4) = 0
munmap(0x7fa9766a0000, 8192) = 0
mmap(NULL, 4096, PROT_READ|PROT_WRITE,
```

```
MAP_PRIVATE|MAP_ANONYMOUS, -1, 0) = 0x7fa9766a1000
openat(AT_FDCWD, "/proc/self/maps", O_RDONLY) = 4
read(4, "20000000000-100000000000 ---p 00"..., 4096) = 4004
read(4, "7fa977730000-7fa977734000 r--p 0"..., 92) = 92
close(4) = 0
munmap(0x7fa9766a1000, 4096) = 0
mmap(NULL, 8192, PROT_READ|PROT_WRITE,
MAP_PRIVATE|MAP_ANONYMOUS, -1, 0) = 0x7fa9766a0000
openat(AT_FDCWD, "/proc/self/maps", O_RDONLY) = 4
read(4, "20000000000-100000000000 ---p 00"..., 8192) = 4071
read(4, "7fa977734000-7fa97773c000 rw-p 0"..., 4121) = 1055
read(4, "", 3066) = 0
close(4) = 0
mmap(0x1b01b39f4000, 8192, PROT_READ, MAP_PRIVATE|MAP_FIXED, 3, 0) =
0x1b01b39f4000
mmap(0x2f52ee598000, 294912, PROT_READ, MAP_PRIVATE|MAP_FIXED, 3, 0) =
0x2f52ee598000
mmap(0x2f52ee60c000, 524288, PROT_READ, MAP_PRIVATE|MAP_FIXED, 3, 0) =
0x2f52ee60c000
mmap(0x2f52ee68c000, 516096, PROT_READ, MAP_PRIVATE|MAP_FIXED, 3, 0) =
0x2f52ee68c000
mmap(0x2f52ee80e000, 524288, PROT_READ, MAP_PRIVATE|MAP_FIXED, 3, 0) =
0x2f52ee80e000
mmap(0x2f52ee88e000, 524288, PROT_READ, MAP_PRIVATE|MAP_FIXED, 3, 0) =
0x2f52ee88e000
mmap(0x2f52ee90e000, 524288, PROT_READ, MAP_PRIVATE|MAP_FIXED, 3, 0) =
0x2f52ee90e000
mmap(0x2f52ee98e000, 524288, PROT_READ, MAP_PRIVATE|MAP_FIXED, 3, 0) =
0x2f52ee98e000
mmap(0x2f52eea0e000, 524288, PROT_READ, MAP_PRIVATE|MAP_FIXED, 3, 0) =
```

0x2f52eea0e000

mmap(0x2f52eea8e000, 524288, PROT_READ, MAP_PRIVATE|MAP_FIXED, 3, 0) =
0x2f52eea8e000

mmap(0x2f52eeb0e000, 65536, PROT_READ, MAP_PRIVATE|MAP_FIXED, 3, 0) =
0x2f52eeb0e000

mmap(0x2f52eec56000, 524288, PROT_READ, MAP_PRIVATE|MAP_FIXED, 3, 0) =
0x2f52eec56000

mmap(0x2f52eecd6000, 524288, PROT_READ, MAP_PRIVATE|MAP_FIXED, 3, 0) =
0x2f52eecd6000

mmap(0x2f52eed56000, 524288, PROT_READ, MAP_PRIVATE|MAP_FIXED, 3, 0) =
0x2f52eed56000

mmap(0x2f52eedd6000, 139264, PROT_READ, MAP_PRIVATE|MAP_FIXED, 3, 0) =
0x2f52eedd6000

mmap(0x2f52f0d20000, 352256, PROT_READ, MAP_PRIVATE|MAP_FIXED, 3, 0) =
0x2f52f0d20000

close(3) = 0

munmap(0x7fa9766a0000, 8192) = 0

munmap(0x7fa976620000, 524288) = 0

sigaltstack(NULL, {ss_sp=NULL, ss_flags=SS_DISABLE, ss_size=0}) = 0

mmap(NULL, 32768, PROT_READ|PROT_WRITE,
MAP_PRIVATE|MAP_ANONYMOUS, -1, 0) = 0x7fa97669a000
sigaltstack({ss_sp=0x7fa97669a000, ss_flags=0, ss_size=32768}, NULL) = 0

rt_sigaction(SIGSEGV, {sa_handler=0x7fa97769af20, sa_mask=[],
sa_flags=SA_RESTORER|SA_ONSTACK|SA_NODEFER|SA_SIGINFO,
sa_restorer=0x7fa977424320}, NULL, 8) = 0

rt_sigaction(SIGBUS, {sa_handler=0x7fa97769af20, sa_mask=[],
sa_flags=SA_RESTORER|SA_ONSTACK|SA_NODEFER|SA_SIGINFO,
sa_restorer=0x7fa977424320}, NULL, 8) = 0

rt_sigaction(SIGFPE, {sa_handler=0x7fa97769af20, sa_mask=[],
sa_flags=SA_RESTORER|SA_ONSTACK|SA_NODEFER|SA_SIGINFO,

```
sa_restorer=0x7fa977424320}, NULL, 8) = 0
mmap(NULL, 2097152, PROT_READ|PROT_WRITE,
MAP_PRIVATE|MAP_ANONYMOUS, -1, 0) = 0x7fa975f72000
munmap(0x7fa975f72000, 581632)      = 0
munmap(0x7fa976100000, 466944)     = 0
mmap(NULL, 2097152, PROT_READ|PROT_WRITE,
MAP_PRIVATE|MAP_ANONYMOUS, -1, 0) = 0x7fa975e00000
munmap(0x7fa975f00000, 1048576)    = 0
mmap(NULL, 4096, PROT_READ|PROT_WRITE,
MAP_PRIVATE|MAP_ANONYMOUS, -1, 0) = 0x7fa976699000
mmap(NULL, 2097152, PROT_READ|PROT_WRITE,
MAP_PRIVATE|MAP_ANONYMOUS, -1, 0) = 0x7fa975c00000
munmap(0x7fa975d00000, 1048576)    = 0
mmap(NULL, 4096, PROT_READ|PROT_WRITE,
MAP_PRIVATE|MAP_ANONYMOUS, -1, 0) = 0x7fa976698000
mmap(NULL, 1703936, PROT_READ|PROT_WRITE,
MAP_PRIVATE|MAP_ANONYMOUS, -1, 0) = 0x7fa975a60000
mmap(NULL, 4096, PROT_READ|PROT_WRITE,
MAP_PRIVATE|MAP_ANONYMOUS, -1, 0) = 0x7fa976697000
mmap(NULL, 2097152, PROT_READ|PROT_WRITE,
MAP_PRIVATE|MAP_ANONYMOUS, -1, 0) = 0x7fa975860000
munmap(0x7fa975860000, 655360)     = 0
munmap(0x7fa975a00000, 393216)     = 0
munmap(0x7fa976697000, 4096)       = 0
mmap(NULL, 2097152, PROT_READ|PROT_WRITE,
MAP_PRIVATE|MAP_ANONYMOUS, -1, 0) = 0x7fa975700000
munmap(0x7fa975800000, 1048576)    = 0
mmap(NULL, 2097152, PROT_READ|PROT_WRITE,
MAP_PRIVATE|MAP_ANONYMOUS, -1, 0) = 0x7fa975500000
munmap(0x7fa975600000, 1048576)    = 0
```

```

munmap(0x7fa976698000, 4096)      = 0
mmap(NULL, 4096, PROT_READ|PROT_WRITE,
MAP_PRIVATE|MAP_ANONYMOUS, -1, 0) = 0x7fa976698000
mmap(NULL, 65536, PROT_READ|PROT_WRITE,
MAP_PRIVATE|MAP_ANONYMOUS, -1, 0) = 0x7fa976688000
mmap(NULL, 8192, PROT_READ|PROT_WRITE,
MAP_PRIVATE|MAP_ANONYMOUS, -1, 0) = 0x7fa976686000
mmap(NULL, 2097152, PROT_READ|PROT_WRITE,
MAP_PRIVATE|MAP_ANONYMOUS, -1, 0) = 0x7fa975300000
munmap(0x7fa975400000, 1048576)    = 0
mmap(NULL, 4096, PROT_READ|PROT_WRITE,
MAP_PRIVATE|MAP_ANONYMOUS, -1, 0) = 0x7fa976685000
gettid()                          = 63073
mmap(NULL, 524288, PROT_READ|PROT_WRITE,
MAP_PRIVATE|MAP_ANONYMOUS|MAP_NORESERVE, -1, 0) = 0x7fa975f80000
madvise(0x7fa975f80000, 524288, MADV_NOHUGEPAGE) = 0
mmap(NULL, 2097152, PROT_READ|PROT_WRITE,
MAP_PRIVATE|MAP_ANONYMOUS, -1, 0) = 0x7fa975100000
munmap(0x7fa975200000, 1048576)    = 0
mmap(NULL, 2097152, PROT_READ|PROT_WRITE,
MAP_PRIVATE|MAP_ANONYMOUS, -1, 0) = 0x7fa974f00000
munmap(0x7fa975000000, 1048576)    = 0
prlimit64(0, RLIMIT_STACK, NULL, {rlim_cur=8192*1024,
rlim_max=RLIM64_INFINITY}) = 0
mmap(NULL, 4096, PROT_READ|PROT_WRITE,
MAP_PRIVATE|MAP_ANONYMOUS, -1, 0) = 0x7fa976684000
openat(AT_FDCWD, "/proc/self/maps", O_RDONLY) = 3
read(3, "200000000000-1000000000000 ---p 00"..., 4096) = 4089
read(3, "7fa975f", 7)                  = 7
close(3)                          = 0

```

```
munmap(0x7fa976684000, 4096)      = 0
mmap(NULL, 8192, PROT_READ|PROT_WRITE,
MAP_PRIVATE|MAP_ANONYMOUS, -1, 0) = 0x7fa976683000
openat(AT_FDCWD, "/proc/self/maps", O_RDONLY) = 3
read(3, "200000000000-1000000000000 ---p 00"..., 8192) = 4089
read(3, "7fa975f80000-7fa976000000 rw-p 0"..., 4103) = 3566
read(3, "", 537)                  = 0
close(3)                          = 0
munmap(0x7fa97661d000, 8192)      = 0
mmap(NULL, 4096, PROT_READ|PROT_WRITE,
MAP_PRIVATE|MAP_ANONYMOUS, -1, 0) = 0x7fa976682000
openat(AT_FDCWD, "/proc/self/maps", O_RDONLY) = 3
read(3, "200000000000-1000000000000 ---p 00"..., 4096) = 4089
read(3, "7fa975f", 7)             = 7
close(3)                          = 0
munmap(0x7fa976682000, 4096)      = 0
mmap(NULL, 8192, PROT_READ|PROT_WRITE,
MAP_PRIVATE|MAP_ANONYMOUS, -1, 0) = 0x7fa976681000
openat(AT_FDCWD, "/proc/self/maps", O_RDONLY) = 3
read(3, "200000000000-1000000000000 ---p 00"..., 8192) = 4089
read(3, "7fa975f80000-7fa976000000 rw-p 0"..., 4103) = 3566
read(3, "", 537)                  = 0
close(3)                          = 0
munmap(0x7fa976681000, 8192)      = 0
mmap(NULL, 262144, PROT_READ|PROT_WRITE,
MAP_PRIVATE|MAP_ANONYMOUS, -1, 0) = 0x7fa976643000
mmap(0x728000000000, 65536, PROT_READ|PROT_WRITE,
MAP_PRIVATE|MAP_FIXED|MAP_ANONYMOUS, -1, 0) = 0x728000000000
mmap(0x728380000000, 65536, PROT_READ|PROT_WRITE,
MAP_PRIVATE|MAP_FIXED|MAP_ANONYMOUS, -1, 0) = 0x728380000000
```

```

mmap(NULL, 65536, PROT_READ|PROT_WRITE,
MAP_PRIVATE|MAP_ANONYMOUS, -1, 0) = 0x7fa976633000
mmap(NULL, 1048576, PROT_READ|PROT_WRITE,
MAP_PRIVATE|MAP_ANONYMOUS, -1, 0) = 0x7fa975d00000
mmap(NULL, 8388608, PROT_READ|PROT_WRITE,
MAP_PRIVATE|MAP_ANONYMOUS|MAP_NORESERVE, -1, 0) = 0x7fa974700000
mmap(NULL, 90112, PROT_READ|PROT_WRITE,
MAP_PRIVATE|MAP_ANONYMOUS, -1, 0) = 0x7fa976609000
clock_gettime(CLOCK_PROCESS_CPUTIME_ID, {tv_sec=0, tv_nsec=24321400}) = 0
rt_sigprocmask(SIG_SETMASK, ~[SYS RT_1], [], 8) = 0
rt_sigaction(SIGRT_1, {sa_handler=0x7fa977478520, sa_mask=[],
sa_flags=SA_RESTORER|SA_ONSTACK|SA_RESTART|SA_SIGINFO,
sa_restorer=0x7fa977424320}, NULL, 8) = 0
rt_sigprocmask(SIG_UNBLOCK, [RTMIN RT_1], NULL, 8) = 0
mmap(NULL, 8392704, PROT_NONE,
MAP_PRIVATE|MAP_ANONYMOUS|MAP_STACK, -1, 0) = 0x7fa973eff000
mprotect(0x7fa973f00000, 8388608, PROT_READ|PROT_WRITE) = 0
mmap(0x724400000000, 65536, PROT_READ|PROT_WRITE,
MAP_PRIVATE|MAP_FIXED|MAP_ANONYMOUS, -1, 0) = 0x724400000000
mmap(0x724780000000, 65536, PROT_READ|PROT_WRITE,
MAP_PRIVATE|MAP_FIXED|MAP_ANONYMOUS, -1, 0) = 0x724780000000
rt_sigprocmask(SIG_BLOCK, ~[], ~[KILL STOP SYS RTMIN RT_1], 8) = 0
clone3({flags=CLONE_VM|CLONE_FS|CLONE_FILES|CLONE_SIGHAND|CLON
E_THREAD|CLONE_SYSVSEM|CLONE_SETTLS|CLONE_PARENT_SETTID|C
LONE_CHILD_CLEARTID, child_tid=0x7fa9746ff990, parent_tid=0x7fa9746ff990,
exit_signal=0, stack=0x7fa973eff000, stack_size=0x7ff7c0, tls=0x7fa9746ff6c0}strace:
Process 63074 attached
=> {parent_tid=[63074]}, 88) = 63074
[pid 63074] rseq(0x7fa9746ffe0, 0x20, 0, 0x53053053 <unfinished ...>
[pid 63073] rt_sigprocmask(SIG_SETMASK, ~[KILL STOP SYS RTMIN RT_1],

```

<unfinished ...>

[pid 63074] <... rseq resumed> = 0

[pid 63073] <... rt_sigprocmask resumed>NULL, 8) = 0

[pid 63074] set_robust_list(0x7fa9746ff9a0, 24 <unfinished ...>

[pid 63073] rt_sigprocmask(SIG_SETMASK, [], <unfinished ...>

[pid 63074] <... set_robust_list resumed>) = 0

[pid 63073] <... rt_sigprocmask resumed>NULL, 8) = 0

[pid 63074] rt_sigprocmask(SIG_SETMASK, ~[KILL STOP SYS RTMIN RT_1],

<unfinished ...>

[pid 63073] mmap(NULL, 8392704, PROT_NONE,

MAP_PRIVATE|MAP_ANONYMOUS|MAP_STACK, -1, 0 <unfinished ...>

[pid 63074] <... rt_sigprocmask resumed>NULL, 8) = 0

[pid 63073] <... mmap resumed> = 0x7fa9736fe000

[pid 63074] gettimeofday(<unfinished ...>

[pid 63073] mprotect(0x7fa9736ff000, 8388608, PROT_READ|PROT_WRITE

<unfinished ...>

[pid 63074] <... gettimeofday resumed>{tv_sec=1733221983, tv_usec=145509}, NULL)
= 0

[pid 63073] <... mprotect resumed> = 0

[pid 63074] nanosleep({tv_sec=0, tv_nsec=100000000}, <unfinished ...>

[pid 63073] rt_sigprocmask(SIG_BLOCK, ~[], [], 8) = 0

[pid 63073]

**clone3({flags=CLONE_VM|CLONE_FS|CLONE_FILES|CLONE_SIGHAND|CLONE_THREAD|CLONE_SYSVSEM|CLONE_SETTLS|CLONE_PARENT_SETTID|CLONE_CHILD_CLEARTID, child_tid=0x7fa973efe990, parent_tid=0x7fa973efe990, exit_signal=0, stack=0x7fa9736fe000, stack_size=0x7ff7c0, tls=0x7fa973efe6c0}strace:
Process 63075 attached**

=> {parent_tid=[63075]}, 88) = 63075

[pid 63075] rseq(0x7fa973efefe0, 0x20, 0, 0x53053053 <unfinished ...>

[pid 63073] rt_sigprocmask(SIG_SETMASK, [], <unfinished ...>

```

[pid 63075] <... rseq resumed>      = 0
[pid 63073] <... rt_sigprocmask resumed>NULL, 8) = 0
[pid 63075] set_robust_list(0x7fa973efe9a0, 24 <unfinished ...>
[pid 63073] mmap(NULL, 4096, PROT_READ|PROT_WRITE,
MAP_PRIVATE|MAP_ANONYMOUS, -1, 0 <unfinished ...>
[pid 63075] <... set_robust_list resumed>) = 0
[pid 63073] <... mmap resumed>      = 0x7fa976632000
[pid 63075] rt_sigprocmask(SIG_SETMASK, [], <unfinished ...>
[pid 63073] futex(0x7ffefb57c0a4, FUTEX_WAKE_PRIVATE, 1 <unfinished ...>
[pid 63075] <... rt_sigprocmask resumed>NULL, 8) = 0
[pid 63073] <... futex resumed>      = 0
[pid 63073] futex(0x7ffefb57c0a8, FUTEX_WAIT_PRIVATE, 0, NULL <unfinished
...>
[pid 63075] gettid()                = 63075
[pid 63075] mmap(NULL, 524288, PROT_READ|PROT_WRITE,
MAP_PRIVATE|MAP_ANONYMOUS|MAP_NORESERVE, -1, 0) = 0x7fa975f00000
[pid 63075] madvise(0x7fa975f00000, 524288, MADV_NOHUGEPAGE) = 0
[pid 63075] sched_getaffinity(63075, 32, [0 1 2 3 4 5 6 7 8 9 10 11 12 13 14 15]) = 32
[pid 63075] mmap(0x722800000000, 65536, PROT_READ|PROT_WRITE,
MAP_PRIVATE|MAP_FIXED|MAP_ANONYMOUS, -1, 0) = 0x722800000000
[pid 63075] mmap(0x722b80000000, 65536, PROT_READ|PROT_WRITE,
MAP_PRIVATE|MAP_FIXED|MAP_ANONYMOUS, -1, 0) = 0x722b80000000
[pid 63075] mmap(NULL, 262144, PROT_READ|PROT_WRITE,
MAP_PRIVATE|MAP_ANONYMOUS, -1, 0) = 0x7fa976132000
[pid 63075] mmap(0x2f52e6dff000, 16756736, PROT_READ|PROT_WRITE,
MAP_PRIVATE|MAP_FIXED|MAP_ANONYMOUS|MAP_NORESERVE, -1, 0) =
0x2f52e6dff000
[pid 63075] madvise(0x2f52e6dff000, 16756736, MADV_NOHUGEPAGE) = 0
[pid 63075] futex(0x7ffefb57c0a8, FUTEX_WAKE_PRIVATE, 1 <unfinished ...>
[pid 63073] <... futex resumed>      = 0

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[pid 63075] <... futex resumed>      = 1
[pid 63073] futex(0x7fa973efe990,
FUTEX_WAIT_BITSET|FUTEX_CLOCK_REALTIME, 63075, NULL,
FUTEX_BITSET_MATCH_ANY <unfinished ...>
[pid 63075] mmap(NULL, 8392704, PROT_NONE,
MAP_PRIVATE|MAP_ANONYMOUS|MAP_STACK, -1, 0) = 0x7fa972efd000
[pid 63075] mprotect(0x7fa972efe000, 8388608, PROT_READ|PROT_WRITE) = 0
[pid 63075] mmap(0x724400010000, 65536, PROT_READ|PROT_WRITE,
MAP_PRIVATE|MAP_FIXED|MAP_ANONYMOUS, -1, 0) = 0x724400010000
[pid 63075] rt_sigprocmask(SIG_BLOCK, ~[], [], 8) = 0
[pid 63075]
clone3({flags=CLONE_VM|CLONE_FS|CLONE_FILES|CLONE_SIGHAND|CLON
E_THREAD|CLONE_SYSVSEM|CLONE_SETTLS|CLONE_PARENT_SETTID|C
LONE_CHILD_CLEARTID, child_tid=0x7fa9736fd990, parent_tid=0x7fa9736fd990,
exit_signal=0, stack=0x7fa972efd000, stack_size=0x7ff7c0,
tls=0x7fa9736fd6c0}strace: Process 63076 attached
<unfinished ...>
[pid 63076] rseq(0x7fa9736fdfe0, 0x20, 0, 0x53053053 <unfinished ...>
[pid 63075] <... clone3 resumed> => {parent_tid=[63076]}, 88) = 63076
[pid 63076] <... rseq resumed>      = 0
[pid 63075] rt_sigprocmask(SIG_SETMASK, [], <unfinished ...>
[pid 63076] set_robust_list(0x7fa9736fd9a0, 24 <unfinished ...>
[pid 63075] <... rt_sigprocmask resumed>NULL, 8) = 0
[pid 63076] <... set_robust_list resumed>) = 0
[pid 63075] futex(0x7fa973efd524, FUTEX_WAKE_PRIVATE, 1 <unfinished ...>
[pid 63076] rt_sigprocmask(SIG_SETMASK, [], <unfinished ...>
[pid 63075] <... futex resumed>      = 0
[pid 63076] <... rt_sigprocmask resumed>NULL, 8) = 0
[pid 63075] futex(0x7fa973efd528, FUTEX_WAIT_PRIVATE, 0, NULL <unfinished
...>

```

```

[pid 63076] getpid()                = 63076
[pid 63076] mmap(NULL, 524288, PROT_READ|PROT_WRITE,
MAP_PRIVATE|MAP_ANONYMOUS|MAP_NORESERVE, -1, 0) = 0x7fa975880000
[pid 63076] madvise(0x7fa975880000, 524288, MADV_NOHUGEPAGE) = 0
[pid 63076] sched_getaffinity(63076, 32, [0 1 2 3 4 5 6 7 8 9 10 11 12 13 14 15]) = 32
[pid 63076] mmap(NULL, 262144, PROT_READ|PROT_WRITE,
MAP_PRIVATE|MAP_ANONYMOUS, -1, 0) = 0x7fa975a20000
[pid 63076] mmap(0x2f52e5dfd000, 16756736, PROT_READ|PROT_WRITE,
MAP_PRIVATE|MAP_FIXED|MAP_ANONYMOUS|MAP_NORESERVE, -1, 0) =
0x2f52e5dfd000
[pid 63076] madvise(0x2f52e5dfd000, 16756736, MADV_NOHUGEPAGE) = 0
[pid 63076] futex(0x7fa973efd528, FUTEX_WAKE_PRIVATE, 1) = 1
[pid 63075] <... futex resumed>      = 0
[pid 63075] mmap(NULL, 8392704, PROT_NONE,
MAP_PRIVATE|MAP_ANONYMOUS|MAP_STACK, -1, 0) = 0x7fa9726fc000
[pid 63075] mprotect(0x7fa9726fd000, 8388608, PROT_READ|PROT_WRITE) = 0
[pid 63075] rt_sigprocmask(SIG_BLOCK, ~[], [], 8) = 0
[pid 63076] mmap(NULL, 262144, PROT_READ|PROT_WRITE,
MAP_PRIVATE|MAP_ANONYMOUS, -1, 0 <unfinished ...>
[pid 63075]
clone3({flags=CLONE_VM|CLONE_FS|CLONE_FILES|CLONE_SIGHAND|CLON
E_THREAD|CLONE_SYSVSEM|CLONE_SETTLS|CLONE_PARENT_SETTID|C
LONE_CHILD_CLEARTID, child_tid=0x7fa972efc990, parent_tid=0x7fa972efc990,
exit_signal=0, stack=0x7fa9726fc000, stack_size=0x7ff7c0, tls=0x7fa972efc6c0}
<unfinished ...>
[pid 63076] <... mmap resumed>      = 0x7fa975840000
strace: Process 63077 attached
[pid 63077] rseq(0x7fa972efcfe0, 0x20, 0, 0x53053053 <unfinished ...>
[pid 63075] <... clone3 resumed> => {parent_tid=[63077]}, 88) = 63077
[pid 63077] <... rseq resumed>      = 0

```

[pid 63076] madvise(0x2f52e5dfc000, 16760832, MADV_DONTNEED <unfinished ...>
 [pid 63077] set_robust_list(0x7fa972efc9a0, 24 <unfinished ...>
 [pid 63075] rt_sigprocmask(SIG_SETMASK, [], <unfinished ...>
 [pid 63077] <... set_robust_list resumed>) = 0
 [pid 63076] <... madvise resumed>) = 0
 [pid 63077] rt_sigprocmask(SIG_SETMASK, [], <unfinished ...>
 [pid 63075] <... rt_sigprocmask resumed>NULL, 8) = 0
 [pid 63077] <... rt_sigprocmask resumed>NULL, 8) = 0
 [pid 63076] madvise(0x2f52e6df9000, 12288, MADV_DONTNEED <unfinished ...>
[pid 63075] futex(0x7fa973efd524, FUTEX_WAKE_PRIVATE, 1 <unfinished ...>
 [pid 63077] getpid(<unfinished ...>
 [pid 63076] <... madvise resumed>) = 0
 [pid 63077] <... getpid resumed>) = 63077
 [pid 63075] <... futex resumed>) = 0
 [pid 63077] mmap(NULL, 524288, PROT_READ|PROT_WRITE,
 MAP_PRIVATE|MAP_ANONYMOUS|MAP_NORESERVE, -1, 0 <unfinished ...>
 [pid 63076] munmap(0x7fa975880000, 524288 <unfinished ...>
 [pid 63077] <... mmap resumed>) = 0x7fa975680000
[pid 63075] futex(0x7fa973efd528, FUTEX_WAIT_PRIVATE, 0, NULL <unfinished
...>
 [pid 63077] madvise(0x7fa975680000, 524288, MADV_NOHUGEPAGE <unfinished ...>
 [pid 63076] <... munmap resumed>) = 0
 [pid 63077] <... madvise resumed>) = 0
[pid 63076] futex(0x7fa9781505a8, FUTEX_WAIT_PRIVATE, 0, NULL <unfinished
...>
[pid 63077] futex(0x7fa9781505a8, FUTEX_WAKE_PRIVATE, 1 <unfinished ...>
 [pid 63076] <... futex resumed>) = -1 EAGAIN (Resource temporarily unavailable)
 [pid 63077] <... futex resumed>) = 0
 [pid 63076] clock_gettime(CLOCK_MONOTONIC, <unfinished ...>
[pid 63077] futex(0x730000000108, FUTEX_WAIT_PRIVATE, 0, NULL <unfinished

...>

[pid 63076] <... clock_gettime resumed>{tv_sec=14139, tv_nsec=143820917}) = 0

[pid 63076] futex(0x730000000108, FUTEX_WAKE_PRIVATE, 1 <unfinished ...>

[pid 63077] <... futex resumed>) = 0

[pid 63076] <... futex resumed>) = 1

[pid 63077] sched_getaffinity(63077, 32, <unfinished ...>

[pid 63076] clock_gettime(CLOCK_MONOTONIC, <unfinished ...>

[pid 63077] <... sched_getaffinity resumed>[0 1 2 3 4 5 6 7 8 9 10 11 12 13 14 15]) = 32

[pid 63076] <... clock_gettime resumed>{tv_sec=14139, tv_nsec=144244974}) = 0

[pid 63077] futex(0x730000000508, FUTEX_WAIT_PRIVATE, 0, NULL <unfinished ...>

[pid 63076] futex(0x730000000508, FUTEX_WAKE_PRIVATE, 1 <unfinished ...>

[pid 63077] <... futex resumed>) = -1 EAGAIN (Resource temporarily unavailable)

[pid 63076] <... futex resumed>) = 0

[pid 63077] mmap(NULL, 262144, PROT_READ|PROT_WRITE,
MAP_PRIVATE|MAP_ANONYMOUS, -1, 0 <unfinished ...>

[pid 63076] rt_sigprocmask(SIG_BLOCK, ~[RT_1], <unfinished ...>

[pid 63077] <... mmap resumed>) = 0x7fa9758c0000

[pid 63076] <... rt_sigprocmask resumed>NULL, 8) = 0

[pid 63077] mmap(0x2f52e4dfb000, 16756736, PROT_READ|PROT_WRITE,
MAP_PRIVATE|MAP_FIXED|MAP_ANONYMOUS|MAP_NORESERVE, -1, 0
<unfinished ...>

[pid 63076] madvise(0x7fa972efd000, 8368128, MADV_DONTNEED <unfinished ...>

[pid 63077] <... mmap resumed>) = 0x2f52e4dfb000

[pid 63076] <... madvise resumed>) = 0

[pid 63077] madvise(0x2f52e4dfb000, 16756736, MADV_NOHUGEPAGE <unfinished ...>

[pid 63076] exit(0 <unfinished ...>

[pid 63077] <... madvise resumed>) = 0

[pid 63076] <... exit resumed>) = ?

[pid 63077] futex(0x7fa973efd528, FUTEX_WAKE_PRIVATE, 1 <unfinished ...>

[pid 63076] +++ exited with 0 +++

[pid 63077] <... futex resumed> = 1

[pid 63075] <... futex resumed> = 0

[pid 63075] futex(0x7fa972efc990,

FUTEX_WAIT_BITSET|FUTEX_CLOCK_REALTIME, 63077, NULL,

FUTEX_BITSET_MATCH_ANY <unfinished ...>

[pid 63077] mmap(NULL, 262144, PROT_READ|PROT_WRITE,

MAP_PRIVATE|MAP_ANONYMOUS, -1, 0) = 0x7fa975880000

[pid 63077] madvise(0x2f52e4dfa000, 16760832, MADV_DONTNEED) = 0

[pid 63077] madvise(0x2f52e5df7000, 12288, MADV_DONTNEED) = 0

[pid 63077] munmap(0x7fa975680000, 524288) = 0

[pid 63077] clock_gettime(CLOCK_MONOTONIC, {tv_sec=14139,
tv_nsec=146871807}) = 0

[pid 63077] clock_gettime(CLOCK_MONOTONIC, {tv_sec=14139,
tv_nsec=147016816}) = 0

[pid 63077] rt_sigprocmask(SIG_BLOCK, ~[RT_1], NULL, 8) = 0

[pid 63077] madvise(0x7fa9726fc000, 8368128, MADV_DONTNEED) = 0

[pid 63077] exit(0) = ?

[pid 63077] +++ exited with 0 +++

[pid 63075] <... futex resumed> = 0

[pid 63075] madvise(0x2f52e6dfe000, 16760832, MADV_DONTNEED) = 0

[pid 63075] madvise(0x2f52e7dfb000, 12288, MADV_DONTNEED) = 0

[pid 63075] munmap(0x7fa975f00000, 524288) = 0

[pid 63075] clock_gettime(CLOCK_MONOTONIC, {tv_sec=14139,
tv_nsec=148361079}) = 0

[pid 63075] clock_gettime(CLOCK_MONOTONIC, {tv_sec=14139,
tv_nsec=148480782}) = 0

[pid 63075] clock_gettime(CLOCK_MONOTONIC, {tv_sec=14139,
tv_nsec=148643437}) = 0

```

[pid 63075] rt_sigprocmask(SIG_BLOCK, ~[RT_1], NULL, 8) = 0
[pid 63075] madvise(0x7fa9736fe000, 8368128, MADV_DONTNEED) = 0
[pid 63075] exit(0) = ?
[pid 63073] <... futex resumed> = 0
[pid 63075] +++ exited with 0 +++
[pid 63073] clock_gettime(CLOCK_PROCESS_CPUTIME_ID, {tv_sec=0,
tv_nsec=31190600}) = 0
[pid 63073] fstat(1, {st_mode=S_IFCHR|0620, st_rdev=makedev(0x88, 0x5), ...}) = 0
[pid 63073] mmap(0x726000000000, 65536, PROT_READ|PROT_WRITE,
MAP_PRIVATE|MAP_FIXED|MAP_ANONYMOUS, -1, 0) = 0x726000000000
[pid 63073] mmap(0x726380000000, 65536, PROT_READ|PROT_WRITE,
MAP_PRIVATE|MAP_FIXED|MAP_ANONYMOUS, -1, 0) = 0x726380000000
[pid 63073] write(1, "Number of threads requested: 4\n", 31Number of threads requested:
4
) = 31
[pid 63073] write(1, "Total threads created: 3\n", 25Total threads created: 3
) = 25
[pid 63073] write(1, "Time taken: 0.006869 seconds\n", 29Time taken: 0.006869 seconds
) = 29
[pid 63073] exit_group(0) = ?
[pid 63074] <... nanosleep resumed> <unfinished ...> = ?
[pid 63074] +++ exited with 0 +++
+++ exited with 0 +++

```

ali_@LAPTOP-TG8SK2OI:~/OS_2/lab_2/src\$./a.out 1000 1

The value of the requested threads: 1

The total number of threads created: 1

Elapsed time: 0.000007 seconds

ali_@LAPTOP-TG8SK2OI:~/OS_2/lab_2/src\$./a.out 1000 2

The value of the requested threads: 2

The total number of threads created: 1

Elapsed time: 0.000008 seconds

ali_@LAPTOP-TG8SK2OI:~/OS_2/lab_2/src\$./a.out 1000 3

The value of the requested threads: 3

The total number of threads created: 3

Elapsed time: 0.000167 seconds

ali_@LAPTOP-TG8SK2OI:~/OS_2/lab_2/src\$./a.out 1000 4

The value of the requested threads: 4

The total number of threads created: 3

Elapsed time: 0.000118 seconds

ali_@LAPTOP-TG8SK2OI:~/OS_2/lab_2/src\$./a.out 1000 5

The value of the requested threads: 5

The total number of threads created: 5

Elapsed time: 0.000282 seconds

ali_@LAPTOP-TG8SK2OI:~/OS_2/lab_2/src\$./a.out 1000 6

The value of the requested threads: 6

The total number of threads created: 5

Elapsed time: 0.000332 seconds

ali_@LAPTOP-TG8SK2OI:~/OS_2/lab_2/src\$./a.out 1000 7

The value of the requested threads: 7

The total number of threads created: 7

Elapsed time: 0.001866 seconds

ali_@LAPTOP-TG8SK2OI:~/OS_2/lab_2/src\$./a.out 1000 8

The value of the requested threads: 8

The total number of threads created: 7

Elapsed time: 0.001582 seconds

ali_@LAPTOP-TG8SK2OI:~/OS_2/lab_2/src\$./a.out 1000 8

The value of the requested threads: 8

The total number of threads created: 7

Elapsed time: 0.003191 seconds

ali_@LAPTOP-TG8SK2OI:~/OS_2/lab_2/src\$./a.out 1000 20

The value of the requested threads: 20

The total number of threads created: 19

Elapsed time: 0.004720 seconds

ali_@LAPTOP-TG8SK2OI:~/OS_2/lab_2/src\$./a.out 1000 1

The value of the requested threads: 1

The total number of threads created: 1

Elapsed time: 0.000007 seconds

ali_@LAPTOP-TG8SK2OI:~/OS_2/lab_2/src\$ strace ./a.out 1000 1

execve("./a.out", ["./a.out", "1000", "1"], 0x7ffeee63e110 /* 29 vars */) = 0

brk(NULL) = 0x564e10342000

arch_prctl(0x3001 /* ARCH_??? */, 0x7ffcba471670) = -1 EINVAL (Invalid argument)

mmap(NULL, 8192, PROT_READ|PROT_WRITE,

MAP_PRIVATE|MAP_ANONYMOUS, -1, 0) = 0x7f5110500000

access("/etc/ld.so.preload", R_OK) = -1 ENOENT (No such file or directory)

openat(AT_FDCWD, "/etc/ld.so.cache", O_RDONLY|O_CLOEXEC) = 3

newfstatat(3, "", {st_mode=S_IFREG|0644, st_size=24583, ...}, AT_EMPTY_PATH) = 0

mmap(NULL, 24583, PROT_READ, MAP_PRIVATE, 3, 0) = 0x7f51104f9000

close(3) = 0

openat(AT_FDCWD, "/lib/x86_64-linux-gnu/libc.so.6", O_RDONLY|O_CLOEXEC) = 3

read(3, "\177ELF\2\1\1\3\0\0\0\0\0\0\0\3\0>\0\1\0\0\0P\237\2\0\0\0\0"..., 832) = 832

pread64(3, "\6\0\0\04\0\0\0@\0\0\0\0\0\0@\0\0\0\0\0\0@\0\0\0\0\0\0"..., 784, 64) = 784

```
pread64(3, "\4\0\0\0 \0\0\0\5\0\0\0GNU\0\2\0\0\300\4\0\0\0\3\0\0\0\0\0\0"..., 48, 848) =
48
pread64(3,
"\4\0\0\0\24\0\0\0\3\0\0\0GNU\0I\17\357\204\3$\f\221\2039x\324\224\323\236S"..., 68,
896) = 68
newfstatat(3, "", {st_mode=S_IFREG|0755, st_size=2220400, ...}, AT_EMPTY_PATH) =
0
pread64(3, "\6\0\0\0\4\0\0\0@\0\0\0\0\0\0\0@\0\0\0\0\0\0\0@\0\0\0\0\0\0\0"..., 784, 64) =
784
mmap(NULL, 2264656, PROT_READ, MAP_PRIVATE|MAP_DENYWRITE, 3, 0) =
0x7f51102d0000
mprotect(0x7f51102f8000, 2023424, PROT_NONE) = 0
mmap(0x7f51102f8000, 1658880, PROT_READ|PROT_EXEC,
MAP_PRIVATE|MAP_FIXED|MAP_DENYWRITE, 3, 0x28000) = 0x7f51102f8000
mmap(0x7f511048d000, 360448, PROT_READ,
MAP_PRIVATE|MAP_FIXED|MAP_DENYWRITE, 3, 0x1bd000) = 0x7f511048d000
mmap(0x7f51104e6000, 24576, PROT_READ|PROT_WRITE,
MAP_PRIVATE|MAP_FIXED|MAP_DENYWRITE, 3, 0x215000) = 0x7f51104e6000
mmap(0x7f51104ec000, 52816, PROT_READ|PROT_WRITE,
MAP_PRIVATE|MAP_FIXED|MAP_ANONYMOUS, -1, 0) = 0x7f51104ec000
close(3) = 0
mmap(NULL, 12288, PROT_READ|PROT_WRITE,
MAP_PRIVATE|MAP_ANONYMOUS, -1, 0) = 0x7f51102cd000
arch_prctl(ARCH_SET_FS, 0x7f51102cd740) = 0
set_tid_address(0x7f51102cda10) = 9241
set_robust_list(0x7f51102cda20, 24) = 0
rseq(0x7f51102ce0e0, 0x20, 0, 0x53053053) = 0
mprotect(0x7f51104e6000, 16384, PROT_READ) = 0
mprotect(0x564e10260000, 4096, PROT_READ) = 0
mprotect(0x7f511053a000, 8192, PROT_READ) = 0
```

```

prlimit64(0, RLIMIT_STACK, NULL, {rlim_cur=8192*1024,
rlim_max=RLIM64_INFINITY}) = 0
munmap(0x7f51104f9000, 24583)      = 0
getrandom("\xe7\x4a\x10\xcd\x12\x7f\x45\xf3", 8, GRND_NONBLOCK) = 8
brk(NULL)                        = 0x564e10342000
brk(0x564e10363000)              = 0x564e10363000
clock_gettime(CLOCK_PROCESS_CPUTIME_ID, {tv_sec=0, tv_nsec=5723400}) = 0
clock_gettime(CLOCK_PROCESS_CPUTIME_ID, {tv_sec=0, tv_nsec=5794500}) = 0
newfstatat(1, "", {st_mode=S_IFCHR|0620, st_rdev=makedev(0x88, 0x3), ...},
AT_EMPTY_PATH) = 0
write(1, "The value of the requested threa"..., 38The value of the requested threads: 1
) = 38
write(1, "The total number of threads crea"..., 39The total number of threads created: 1
) = 39
write(1, "Elapsed time: 0.000071 seconds\n", 31Elapsed time: 0.000071 seconds
) = 31
exit_group(0)                    = ?
+++ exited with 0 +++

```

```

ali_@LAPTOP-TG8SK2OI:~/OS_2/lab_2/src$ strace -f ./a.out 1000 2
execve("./a.out", [ "./a.out", "1000", "2"], 0x7fffb943e1a8 /* 29 vars */) = 0
brk(NULL)                        = 0x55d617dae000
arch_prctl(0x3001 /* ARCH_??? */, 0x7ffee4f13f70) = -1 EINVAL (Invalid argument)
mmap(NULL, 8192, PROT_READ|PROT_WRITE,
MAP_PRIVATE|MAP_ANONYMOUS, -1, 0) = 0x7ffa7b805000
access("/etc/ld.so.preload", R_OK) = -1 ENOENT (No such file or directory)
openat(AT_FDCWD, "/etc/ld.so.cache", O_RDONLY|O_CLOEXEC) = 3

```

```

newfstatat(3, "", {st_mode=S_IFREG|0644, st_size=24583, ...}, AT_EMPTY_PATH) = 0
mmap(NULL, 24583, PROT_READ, MAP_PRIVATE, 3, 0) = 0x7ffa7b7fe000
close(3) = 0
openat(AT_FDCWD, "/lib/x86_64-linux-gnu/libc.so.6", O_RDONLY|O_CLOEXEC) = 3
read(3, "\177ELF\2\1\1\3\0\0\0\0\0\0\0\3\0>\0\1\0\0\0P\237\2\0\0\0\0"..., 832) = 832
pread64(3, "\6\0\0\0\4\0\0\0@\0\0\0\0\0\0\0@\0\0\0\0\0\0\0@\0\0\0\0\0\0\0"..., 784, 64) =
784
pread64(3, "\4\0\0\0 \0\0\0\5\0\0\0GNU\0\2\0\0\300\4\0\0\0\3\0\0\0\0\0\0"..., 48, 848) =
48
pread64(3,
"\4\0\0\0\24\0\0\0\3\0\0\0GNU\0I\17\357\204\3$\f\221\2039x\324\224\323\236S"..., 68,
896) = 68
newfstatat(3, "", {st_mode=S_IFREG|0755, st_size=2220400, ...}, AT_EMPTY_PATH) =
0
pread64(3, "\6\0\0\0\4\0\0\0@\0\0\0\0\0\0\0@\0\0\0\0\0\0\0@\0\0\0\0\0\0\0"..., 784, 64) =
784
mmap(NULL, 2264656, PROT_READ, MAP_PRIVATE|MAP_DENYWRITE, 3, 0) =
0x7ffa7b5d5000
mprotect(0x7ffa7b5fd000, 2023424, PROT_NONE) = 0
mmap(0x7ffa7b5fd000, 1658880, PROT_READ|PROT_EXEC,
MAP_PRIVATE|MAP_FIXED|MAP_DENYWRITE, 3, 0x28000) = 0x7ffa7b5fd000
mmap(0x7ffa7b792000, 360448, PROT_READ,
MAP_PRIVATE|MAP_FIXED|MAP_DENYWRITE, 3, 0x1bd000) = 0x7ffa7b792000
mmap(0x7ffa7b7eb000, 24576, PROT_READ|PROT_WRITE,
MAP_PRIVATE|MAP_FIXED|MAP_DENYWRITE, 3, 0x215000) = 0x7ffa7b7eb000
mmap(0x7ffa7b7f1000, 52816, PROT_READ|PROT_WRITE,
MAP_PRIVATE|MAP_FIXED|MAP_ANONYMOUS, -1, 0) = 0x7ffa7b7f1000
close(3) = 0
mmap(NULL, 12288, PROT_READ|PROT_WRITE,
MAP_PRIVATE|MAP_ANONYMOUS, -1, 0) = 0x7ffa7b5d2000

```

```

arch_prctl(ARCH_SET_FS, 0x7ffa7b5d2740) = 0
set_tid_address(0x7ffa7b5d2a10)      = 9272
set_robust_list(0x7ffa7b5d2a20, 24)  = 0
rseq(0x7ffa7b5d30e0, 0x20, 0, 0x53053053) = 0
mprotect(0x7ffa7b7eb000, 16384, PROT_READ) = 0
mprotect(0x55d617a59000, 4096, PROT_READ) = 0
mprotect(0x7ffa7b83f000, 8192, PROT_READ) = 0
prlimit64(0, RLIMIT_STACK, NULL, {rlim_cur=8192*1024,
rlim_max=RLIM64_INFINITY}) = 0
munmap(0x7ffa7b7fe000, 24583)        = 0
getrandom("\x99\xce\xe9\xf5\xfc\x7d\xaa\x16", 8, GRND_NONBLOCK) = 8
brk(NULL)                            = 0x55d617dae000
brk(0x55d617dcf000)                  = 0x55d617dcf000
clock_gettime(CLOCK_PROCESS_CPUTIME_ID, {tv_sec=0, tv_nsec=5323500}) = 0
clock_gettime(CLOCK_PROCESS_CPUTIME_ID, {tv_sec=0, tv_nsec=5434400}) = 0
newfstatat(1, "", {st_mode=S_IFCHR|0620, st_rdev=makedev(0x88, 0x3), ...},
AT_EMPTY_PATH) = 0
write(1, "The value of the requested threa"..., 38The value of the requested threads: 2
) = 38
write(1, "The total number of threads crea"..., 39The total number of threads created: 1
) = 39
write(1, "Elapsed time: 0.000111 seconds\n", 31Elapsed time: 0.000111 seconds
) = 31
exit_group(0)                        = ?
+++ exited with 0 +++

```

```

ali_@LAPTOP-TG8SK2OI:~/OS_2/lab_2/src$ strace -f ./a.out 1000 8
execve("./a.out", ["/a.out", "1000", "8"], 0x7ffc2aa7368 /* 29 vars */) = 0

```

```
brk(NULL) = 0x5581c09e0000
arch_prctl(0x3001 /* ARCH_??? */, 0x7fff31d85290) = -1 EINVAL (Invalid argument)
mmap(NULL, 8192, PROT_READ|PROT_WRITE,
MAP_PRIVATE|MAP_ANONYMOUS, -1, 0) = 0x7f2d82fb8000
access("/etc/ld.so.preload", R_OK) = -1 ENOENT (No such file or directory)
openat(AT_FDCWD, "/etc/ld.so.cache", O_RDONLY|O_CLOEXEC) = 3
newfstatat(3, "", {st_mode=S_IFREG|0644, st_size=24583, ...}, AT_EMPTY_PATH) = 0
mmap(NULL, 24583, PROT_READ, MAP_PRIVATE, 3, 0) = 0x7f2d82fb1000
close(3) = 0
openat(AT_FDCWD, "/lib/x86_64-linux-gnu/libc.so.6", O_RDONLY|O_CLOEXEC) = 3
read(3, "\177ELF\2\1\1\3\0\0\0\0\0\0\0\3\0>\0\1\0\0\0P\237\2\0\0\0\0"..., 832) = 832
pread64(3, "\6\0\0\04\0\0\0@\0\0\0\0\0\0@\0\0\0\0\0\0@\0\0\0\0\0\0"..., 784, 64) =
784
pread64(3, "\4\0\0\0 \0\0\05\0\0\0GNU\02\0\0300\4\0\0\03\0\0\0\0\0\0"..., 48, 848) =
48
pread64(3,
"\4\0\0\024\0\0\03\0\0\0GNU\0I\17\357\204\3$\f221\2039x\324\224\323\236S"..., 68,
896) = 68
newfstatat(3, "", {st_mode=S_IFREG|0755, st_size=2220400, ...}, AT_EMPTY_PATH) =
0
pread64(3, "\6\0\0\04\0\0\0@\0\0\0\0\0\0@\0\0\0\0\0\0@\0\0\0\0\0\0"..., 784, 64) =
784
mmap(NULL, 2264656, PROT_READ, MAP_PRIVATE|MAP_DENYWRITE, 3, 0) =
0x7f2d82d88000
mprotect(0x7f2d82db0000, 2023424, PROT_NONE) = 0
mmap(0x7f2d82db0000, 1658880, PROT_READ|PROT_EXEC,
MAP_PRIVATE|MAP_FIXED|MAP_DENYWRITE, 3, 0x28000) = 0x7f2d82db0000
mmap(0x7f2d82f45000, 360448, PROT_READ,
MAP_PRIVATE|MAP_FIXED|MAP_DENYWRITE, 3, 0x1bd000) = 0x7f2d82f45000
mmap(0x7f2d82f9e000, 24576, PROT_READ|PROT_WRITE,
```

```

MAP_PRIVATE|MAP_FIXED|MAP_DENYWRITE, 3, 0x215000) = 0x7f2d82f9e000
mmap(0x7f2d82fa4000, 52816, PROT_READ|PROT_WRITE,
MAP_PRIVATE|MAP_FIXED|MAP_ANONYMOUS, -1, 0) = 0x7f2d82fa4000
close(3) = 0
mmap(NULL, 12288, PROT_READ|PROT_WRITE,
MAP_PRIVATE|MAP_ANONYMOUS, -1, 0) = 0x7f2d82d85000
arch_prctl(ARCH_SET_FS, 0x7f2d82d85740) = 0
set_tid_address(0x7f2d82d85a10) = 9309
set_robust_list(0x7f2d82d85a20, 24) = 0
rseq(0x7f2d82d860e0, 0x20, 0, 0x53053053) = 0
mprotect(0x7f2d82f9e000, 16384, PROT_READ) = 0
mprotect(0x5581bf8ba000, 4096, PROT_READ) = 0
mprotect(0x7f2d82ff2000, 8192, PROT_READ) = 0
prlimit64(0, RLIMIT_STACK, NULL, {rlim_cur=8192*1024,
rlim_max=RLIM64_INFINITY}) = 0
munmap(0x7f2d82fb1000, 24583) = 0
getrandom("\xe8\xfe\xdc\x46\xf9\x15\xd8\x2d", 8, GRND_NONBLOCK) = 8
brk(NULL) = 0x5581c09e0000
brk(0x5581c0a01000) = 0x5581c0a01000
clock_gettime(CLOCK_PROCESS_CPUTIME_ID, {tv_sec=0, tv_nsec=4805100}) = 0
rt_sigaction(SIGRT_1, {sa_handler=0x7f2d82e19870, sa_mask=[],
sa_flags=SA_RESTORER|SA_ONSTACK|SA_RESTART|SA_SIGINFO,
sa_restorer=0x7f2d82dca520}, NULL, 8) = 0
rt_sigprocmask(SIG_UNBLOCK, [RTMIN RT_1], NULL, 8) = 0
mmap(NULL, 8392704, PROT_NONE,
MAP_PRIVATE|MAP_ANONYMOUS|MAP_STACK, -1, 0) = 0x7f2d82584000
mprotect(0x7f2d82585000, 8388608, PROT_READ|PROT_WRITE) = 0
rt_sigprocmask(SIG_BLOCK, ~[], [], 8) = 0
clone3({flags=CLONE_VM|CLONE_FS|CLONE_FILES|CLONE_SIGHAND|CLON
E_THREAD|CLONE_SYSVSEM|CLONE_SETTLS|CLONE_PARENT_SETTID|C

```

**LONE_CHILD_CLEARTID, child_tid=0x7f2d82d84910,
parent_tid=0x7f2d82d84910, exit_signal=0, stack=0x7f2d82584000,
stack_size=0x7fff00, tls=0x7f2d82d84640}strace: Process 9310 attached
=> {parent_tid=[9310]}, 88) = 9310**

[pid 9309] rt_sigprocmask(SIG_SETMASK, [], <unfinished ...>
[pid 9310] rseq(0x7f2d82d84fe0, 0x20, 0, 0x53053053 <unfinished ...>
[pid 9309] <... rt_sigprocmask resumed>NULL, 8) = 0
[pid 9310] <... rseq resumed> = 0
[pid 9309] mmap(NULL, 8392704, PROT_NONE,
MAP_PRIVATE|MAP_ANONYMOUS|MAP_STACK, -1, 0 <unfinished ...>
[pid 9310] set_robust_list(0x7f2d82d84920, 24 <unfinished ...>
[pid 9309] <... mmap resumed> = 0x7f2d81d83000
[pid 9310] <... set_robust_list resumed>) = 0
[pid 9309] mprotect(0x7f2d81d84000, 8388608, PROT_READ|PROT_WRITE
<unfinished ...>

[pid 9310] rt_sigprocmask(SIG_SETMASK, [], <unfinished ...>
[pid 9309] <... mprotect resumed> = 0
[pid 9310] <... rt_sigprocmask resumed>NULL, 8) = 0
[pid 9309] rt_sigprocmask(SIG_BLOCK, ~[], <unfinished ...>
[pid 9310] mmap(NULL, 8392704, PROT_NONE,
MAP_PRIVATE|MAP_ANONYMOUS|MAP_STACK, -1, 0 <unfinished ...>
[pid 9309] <... rt_sigprocmask resumed>[], 8) = 0
[pid 9310] <... mmap resumed> = 0x7f2d81582000
[pid 9309]

**clone3({flags=CLONE_VM|CLONE_FS|CLONE_FILES|CLONE_SIGHAND|CLON
E_THREAD|CLONE_SYSVSEM|CLONE_SETTLS|CLONE_PARENT_SETTID|C
LONE_CHILD_CLEARTID, child_tid=0x7f2d82583910,
parent_tid=0x7f2d82583910, exit_signal=0, stack=0x7f2d81d83000,
stack_size=0x7fff00, tls=0x7f2d82583640} <unfinished ...>
[pid 9310] mprotect(0x7f2d81583000, 8388608, PROT_READ|PROT_WRITEstrace:**

Process 9311 attached

) = 0

[pid 9309] <... clone3 resumed> => {parent_tid=[9311]}, 88) = 9311

[pid 9311] rseq(0x7f2d82583fe0, 0x20, 0, 0x53053053 <unfinished ...>

[pid 9310] mmap(NULL, 134217728, PROT_NONE,
MAP_PRIVATE|MAP_ANONYMOUS|MAP_NORESERVE, -1, 0 <unfinished ...>

[pid 9309] rt_sigprocmask(SIG_SETMASK, [], <unfinished ...>

[pid 9311] <... rseq resumed> = 0

[pid 9309] <... rt_sigprocmask resumed>NULL, 8) = 0

[pid 9310] <... mmap resumed> = 0x7f2d79582000

[pid 9311] set_robust_list(0x7f2d82583920, 24 <unfinished ...>

**[pid 9309] futex(0x7f2d82d84910,
FUTEX_WAIT_BITSET|FUTEX_CLOCK_REALTIME, 9310, NULL,
FUTEX_BITSET_MATCH_ANY <unfinished ...>**

[pid 9311] <... set_robust_list resumed>) = 0

[pid 9310] munmap(0x7f2d79582000, 44556288 <unfinished ...>

[pid 9311] rt_sigprocmask(SIG_SETMASK, [], <unfinished ...>

[pid 9310] <... munmap resumed> = 0

[pid 9310] munmap(0x7f2d80000000, 22552576 <unfinished ...>

[pid 9311] <... rt_sigprocmask resumed>NULL, 8) = 0

[pid 9310] <... munmap resumed> = 0

[pid 9310] mprotect(0x7f2d7c000000, 135168, PROT_READ|PROT_WRITE
<unfinished ...>

[pid 9311] mmap(NULL, 8392704, PROT_NONE,
MAP_PRIVATE|MAP_ANONYMOUS|MAP_STACK, -1, 0 <unfinished ...>

[pid 9310] <... mprotect resumed> = 0

[pid 9311] <... mmap resumed> = 0x7f2d80d81000

[pid 9310] rt_sigprocmask(SIG_BLOCK, ~[], <unfinished ...>

[pid 9311] mprotect(0x7f2d80d82000, 8388608, PROT_READ|PROT_WRITE
<unfinished ...>

```

[pid 9310] <... rt_sigprocmask resumed>[], 8) = 0
[pid 9311] <... mprotect resumed>)    = 0
[pid 9310]
clone3({flags=CLONE_VM|CLONE_FS|CLONE_FILES|CLONE_SIGHAND|CLON
E_THREAD|CLONE_SYSVSEM|CLONE_SETTLS|CLONE_PARENT_SETTID|C
LONE_CHILD_CLEARTID, child_tid=0x7f2d81d82910,
parent_tid=0x7f2d81d82910, exit_signal=0, stack=0x7f2d81582000,
stack_size=0x7fff00, tls=0x7f2d81d82640} <unfinished ...>
[pid 9311] mmap(NULL, 134217728, PROT_NONE,
MAP_PRIVATE|MAP_ANONYMOUS|MAP_NORESERVE, -1, 0strace: Process
9312 attached
) = 0x7f2d74000000
[pid 9312] rseq(0x7f2d81d82fe0, 0x20, 0, 0x53053053 <unfinished ...>
[pid 9311] munmap(0x7f2d78000000, 67108864 <unfinished ...>
[pid 9312] <... rseq resumed>)        = 0
[pid 9310] <... clone3 resumed> => {parent_tid=[9312]}, 88) = 9312
[pid 9312] set_robust_list(0x7f2d81d82920, 24 <unfinished ...>
[pid 9311] <... munmap resumed>)      = 0
[pid 9312] <... set_robust_list resumed>) = 0
[pid 9310] rt_sigprocmask(SIG_SETMASK, [], <unfinished ...>
[pid 9312] rt_sigprocmask(SIG_SETMASK, [], <unfinished ...>
[pid 9311] mprotect(0x7f2d74000000, 135168, PROT_READ|PROT_WRITE
<unfinished ...>
[pid 9312] <... rt_sigprocmask resumed>NULL, 8) = 0
[pid 9310] <... rt_sigprocmask resumed>NULL, 8) = 0
[pid 9312] mmap(0x7f2d78000000, 67108864, PROT_NONE,
MAP_PRIVATE|MAP_ANONYMOUS|MAP_NORESERVE, -1, 0 <unfinished ...>
[pid 9311] <... mprotect resumed>)    = 0
[pid 9312] <... mmap resumed>)       = 0x7f2d78000000
[pid 9310] mmap(NULL, 8392704, PROT_NONE,

```

```

MAP_PRIVATE|MAP_ANONYMOUS|MAP_STACK, -1, 0 <unfinished ...>
[pid 9312] mprotect(0x7f2d78000000, 135168, PROT_READ|PROT_WRITE
<unfinished ...>
[pid 9311] rt_sigprocmask(SIG_BLOCK, ~[], <unfinished ...>
[pid 9312] <... mprotect resumed>    = 0
[pid 9310] <... mmap resumed>       = 0x7f2d80580000
[pid 9311] <... rt_sigprocmask resumed>[], 8) = 0
[pid 9310] mprotect(0x7f2d80581000, 8388608, PROT_READ|PROT_WRITE
<unfinished ...>
[pid 9311]
clone3({flags=CLONE_VM|CLONE_FS|CLONE_FILES|CLONE_SIGHAND|CLON
E_THREAD|CLONE_SYSVSEM|CLONE_SETTLS|CLONE_PARENT_SETTID|C
LONE_CHILD_CLEARTID, child_tid=0x7f2d81581910,
parent_tid=0x7f2d81581910, exit_signal=0, stack=0x7f2d80d81000,
stack_size=0x7fff00, tls=0x7f2d81581640} <unfinished ...>
[pid 9310] <... mprotect resumed>    = 0
strace: Process 9313 attached
[pid 9311] <... clone3 resumed> => {parent_tid=[9313]}, 88) = 9313
[pid 9310] rt_sigprocmask(SIG_BLOCK, ~[], <unfinished ...>
[pid 9313] rseq(0x7f2d81581fe0, 0x20, 0, 0x53053053 <unfinished ...>
[pid 9311] rt_sigprocmask(SIG_SETMASK, [], <unfinished ...>
[pid 9310] <... rt_sigprocmask resumed>[], 8) = 0
[pid 9313] <... rseq resumed>       = 0
[pid 9311] <... rt_sigprocmask resumed>NULL, 8) = 0
[pid 9310]
clone3({flags=CLONE_VM|CLONE_FS|CLONE_FILES|CLONE_SIGHAND|CLON
E_THREAD|CLONE_SYSVSEM|CLONE_SETTLS|CLONE_PARENT_SETTID|C
LONE_CHILD_CLEARTID, child_tid=0x7f2d80d80910,
parent_tid=0x7f2d80d80910, exit_signal=0, stack=0x7f2d80580000,
stack_size=0x7fff00, tls=0x7f2d80d80640} <unfinished ...>

```

```

[pid 9313] set_robust_list(0x7f2d81581920, 24 <unfinished ...>
[pid 9311] mmap(NULL, 8392704, PROT_NONE,
MAP_PRIVATE|MAP_ANONYMOUS|MAP_STACK, -1, 0strace: Process 9314 attached
<unfinished ...>
[pid 9313] <... set_robust_list resumed>) = 0
[pid 9311] <... mmap resumed>      = 0x7f2d737ff000
[pid 9313] rt_sigprocmask(SIG_SETMASK, [], <unfinished ...>
[pid 9314] rseq(0x7f2d80d80fe0, 0x20, 0, 0x53053053 <unfinished ...>
[pid 9311] mprotect(0x7f2d73800000, 8388608, PROT_READ|PROT_WRITE
<unfinished ...>
[pid 9310] <... clone3 resumed> => {parent_tid=[9314]}, 88) = 9314
[pid 9314] <... rseq resumed>      = 0
[pid 9313] <... rt_sigprocmask resumed>NULL, 8) = 0
[pid 9314] set_robust_list(0x7f2d80d80920, 24 <unfinished ...>
[pid 9311] <... mprotect resumed>)  = 0
[pid 9310] rt_sigprocmask(SIG_SETMASK, [], <unfinished ...>
[pid 9312] rt_sigprocmask(SIG_BLOCK, ~[RT_1], NULL, 8) = 0
[pid 9312] madvise(0x7f2d81582000, 8368128, MADV_DONTNEED) = 0
[pid 9312] exit(0)          = ?
[pid 9312] +++ exited with 0 +++
[pid 9311] rt_sigprocmask(SIG_BLOCK, ~[], <unfinished ...>
[pid 9314] <... set_robust_list resumed>) = 0
[pid 9313] rt_sigprocmask(SIG_BLOCK, ~[RT_1], <unfinished ...>
[pid 9310] <... rt_sigprocmask resumed>NULL, 8) = 0
[pid 9314] rt_sigprocmask(SIG_SETMASK, [], <unfinished ...>
[pid 9313] <... rt_sigprocmask resumed>NULL, 8) = 0
[pid 9311] <... rt_sigprocmask resumed>[], 8) = 0
[pid 9314] <... rt_sigprocmask resumed>NULL, 8) = 0
[pid 9313] madvise(0x7f2d80d81000, 8368128, MADV_DONTNEED <unfinished ...>
[pid 9310] futex(0x7f2d80d80910,

```

FUTEX_WAIT_BITSET|FUTEX_CLOCK_REALTIME, 9314, NULL,

FUTEX_BITSET_MATCH_ANY <unfinished ...>

[pid 9314] rt_sigprocmask(SIG_BLOCK, ~[RT_1], <unfinished ...>

[pid 9313] <... madvise resumed>) = 0

[pid 9311]

clone3({flags=CLONE_VM|CLONE_FS|CLONE_FILES|CLONE_SIGHAND|CLONE_THREAD|CLONE_SYSVSEM|CLONE_SETTLS|CLONE_PARENT_SETTID|CLONE_CHILD_CLEARTID, child_tid=0x7f2d73fff910, parent_tid=0x7f2d73fff910, exit_signal=0, stack=0x7f2d737ff000, stack_size=0x7fff00, tls=0x7f2d73fff640}

<unfinished ...>

[pid 9314] <... rt_sigprocmask resumed>NULL, 8) = 0

[pid 9313] exit(0strace: Process 9315 attached

<unfinished ...>

[pid 9314] madvise(0x7f2d80580000, 8368128, MADV_DONTNEED <unfinished ...>

[pid 9313] <... exit resumed>) = ?

[pid 9315] rseq(0x7f2d73ffffe0, 0x20, 0, 0x53053053 <unfinished ...>

[pid 9314] <... madvise resumed>) = 0

[pid 9313] +++ exited with 0 +++

[pid 9311] <... clone3 resumed> => {parent_tid=[9315]}, 88) = 9315

[pid 9315] <... rseq resumed>) = 0

[pid 9314] exit(0 <unfinished ...>

[pid 9315] set_robust_list(0x7f2d73fff920, 24 <unfinished ...>

[pid 9311] rt_sigprocmask(SIG_SETMASK, [], <unfinished ...>

[pid 9315] <... set_robust_list resumed>) = 0

[pid 9314] <... exit resumed>) = ?

[pid 9315] rt_sigprocmask(SIG_SETMASK, [], <unfinished ...>

[pid 9311] <... rt_sigprocmask resumed>NULL, 8) = 0

[pid 9315] <... rt_sigprocmask resumed>NULL, 8) = 0

[pid 9314] +++ exited with 0 +++

[pid 9310] <... futex resumed>) = 0

```

[pid 9315] rt_sigprocmask(SIG_BLOCK, ~[RT_1], <unfinished ...>
[pid 9311] futex(0x7f2d73fff910,
FUTEX_WAIT_BITSET|FUTEX_CLOCK_REALTIME, 9315, NULL,
FUTEX_BITSET_MATCH_ANY <unfinished ...>
[pid 9315] <... rt_sigprocmask resumed>NULL, 8) = 0
[pid 9310] rt_sigprocmask(SIG_BLOCK, ~[RT_1], <unfinished ...>
[pid 9315] madvise(0x7f2d737ff000, 8368128, MADV_DONTNEED <unfinished ...>
[pid 9310] <... rt_sigprocmask resumed>NULL, 8) = 0
[pid 9315] <... madvise resumed>    = 0
[pid 9310] madvise(0x7f2d82584000, 8368128, MADV_DONTNEED <unfinished ...>
[pid 9315] exit(0 <unfinished ...>
[pid 9310] <... madvise resumed>    = 0
[pid 9315] <... exit resumed>      = ?
[pid 9310] exit(0 <unfinished ...>
[pid 9315] +++ exited with 0 +++
[pid 9311] <... futex resumed>      = 0
[pid 9310] <... exit resumed>      = ?
[pid 9311] rt_sigprocmask(SIG_BLOCK, ~[RT_1], <unfinished ...>
[pid 9309] <... futex resumed>      = 0
[pid 9311] <... rt_sigprocmask resumed>NULL, 8) = 0
[pid 9310] +++ exited with 0 +++
[pid 9309] munmap(0x7f2d81582000, 8392704 <unfinished ...>
[pid 9311] madvise(0x7f2d81d83000, 8368128, MADV_DONTNEED <unfinished ...>
[pid 9309] <... munmap resumed>    = 0
[pid 9311] <... madvise resumed>    = 0
[pid 9309] futex(0x7f2d82583910,
FUTEX_WAIT_BITSET|FUTEX_CLOCK_REALTIME, 9311, NULL,
FUTEX_BITSET_MATCH_ANY <unfinished ...>
[pid 9311] exit(0)                  = ?
[pid 9309] <... futex resumed>      = 0

```

```

[pid 9311] +++ exited with 0 +++
munmap(0x7f2d80d81000, 8392704)      = 0
clock_gettime(CLOCK_PROCESS_CPUTIME_ID, {tv_sec=0, tv_nsec=13878300}) = 0
newfstatat(1, "", {st_mode=S_IFCHR|0620, st_rdev=makedev(0x88, 0x3), ...},
AT_EMPTY_PATH) = 0
write(1, "The value of the requested threa"..., 38The value of the requested threads: 8
) = 38
write(1, "The total number of threads crea"..., 39The total number of threads created: 7
) = 39
write(1, "Elapsed time: 0.009073 seconds\n", 31Elapsed time: 0.009073 seconds
) = 31
exit_group(0)                      = ?
+++ exited with 0 +++
ali_@LAPTOP-TG8SK2OI:~/OS_2/lab_2/src$

```

Вывод

В ходе выполнения данной лабораторной работы я научилась создавать программы на языке C, обрабатывающие данные в многопоточном режиме, применять стандартные средства операционной системы для управления потоками и их синхронизации.

В ходе тестирования программы выполняющей многопоточную сортировку слиянием, я обнаружила, что увеличение числа потоков приводит к заметному ускорению выполнения программы при обработке больших массивов данных. Недостаток эффективности происходит из-за использования malloc для выделения памяти для подмассивов. Однако эффективность использования вычислительных ресурсов начинает снижаться при превышении количества потоков,

соответствующего числу логических ядер процессора. Это связано с дополнительными затратами на управление потоками и синхронизацию между ними.